

shop floors can be predicted based on analysing historical data using some well-known machine learning models. The quality of power too can be managed using AI models, which can be programmed in such a way as to provide fluctuation-free and uninterrupted power supply with optimum energy usage. The AI+IoT combination can also be used to maintain near rated voltages around 230V, constantly. The AI system considers many factors to optimise performance.

Managing smart cities using AI has become very popular. If each household can optimise its light usage or save energy while operating the various power consuming gadgets in the house, imagine how much power can be saved across the entire city.

Even in industries, smart energy initiatives along with AI+IoT can help in solving many challenges and optimising power usage.

Media and entertainment: This sector faces a number of challenges in audio, video and image processing, which can be addressed using advanced AI, ML and DL techniques. Volumes of data are being generated in the media and entertainment sector every day, either video or audio based. Transmitting this multimedia data consumes a lot of Internet bandwidth and also hampers the transfer rate.

Some techniques like advanced data compression can be used to make the transfer efficient and faster. IoT+ AI can be used on various data compression methodologies, which can be based on historical data for better prediction and usage. Also, by using AI, only the relevant data of the audio/video combination can be sent out. This selection is calculated independently using advanced AI algorithms, which saves a lot of bandwidth and also increases the transfer rate.

Healthcare: Managing and curing many diseases requires genetic algorithmic solutions. This sector is

one of the key areas in which there is considerable ongoing development and advances in AI (especially DL).

In recent times, 'decision support systems' have been increasingly making use of AI technologies to assist clinicians. After analysing certain reports, faster decisions on diagnosing diseases and their causes can be made with the help of AI techniques. CNN based screening of X-ray images to classify various diseases is picking up. Also, image or non-image data of patients can be fused together to infer the clinical conditions of the patient. For example, in the ongoing COVID-19 pandemic, many AI based techniques are being used to screen patients, based on subtle features.

Manufacturing: Most manufacturing industries rely heavily on combinations of AI+ML+DL technologies. Even Industry 4.0 can be blended with AI technologies, for a smarter output and to better tackle industry-specific challenges.

The challenge of robot and drone co-ordination can be addressed using AI techniques. Both the robot and the drone can be individually controlled or instructed to handle or predict certain outcomes in the manufacturing industry. These can be used for handling or for preventing disaster situations.

Solutions like workflow optimisation and the division of labour (between machines and humans) require advanced AI algorithms and techniques. For OEM vendors of IoT (making the right choice of device at the right time, based on prevailing parameters), the latest AI techniques can be used for deciding which combination of IoT to choose.

Retail: The retail industry is one sector that serves the daily needs of many consumers. Traditional methods of shopping are time consuming and require a lot of human effort, but with combinations of AI + AR/VR and using virtual human-like avatars, multiple shopping experiences can be

provided with real-time simulation of the shopping floor. Consumers get a mesmerising experience even while shopping in the comfort of their homes.

Telecom: Upcoming technologies like 5G are increasingly making use of AI for traffic shaping, congestion control and to ensure committed SLAs. Based on the type of data to be transferred, resources are allocated using AI algorithms (for prediction). The next big challenges like the seamless protocol or bulk transfer of data can be addressed using the AI+5G combination. This will help in solving most of the high-end data transmission challenges.

Automotive: The use of self-driving cars is expected to become more widespread. These autonomous vehicles use many of the AI features for predicting obstacles on the road, reading signboards, gauging the speed of oncoming vehicles and their direction, etc. All these identification processes have to happen in real-time to prevent any accidents.

Many of the latest vehicles are equipped with dashboards with advanced control systems built-in, and prediction and correction units. Also, warning systems are built based on the driver's gestures, behaviour or posture. These will help in preventing many accidents. Figure 1 is a sample signboard, and Figure 2 shows how a computer reads it and makes predictions.

Finance: This is one sector that is continuously evolving and has to handle a lot of customer related challenges, which are not only human effort intensive but also time consuming. For example, when granting loans to certain customers, one has to manually examine their credit histories or backgrounds. But with a combination of AI technologies, client history or credit score data can be fed to machine learning models (AI systems) and later be used to make predictions. These predictive models can be built using historical data sets.